Stanford Health Care Antimicrobial Dosing Reference Guide

This document is also located on the SHC Intranet (http://portal.stanfordmed.org/depts/AntimicrobialStewardshipProgram) and http://bugsanddrugs.stanford.edu ▴ ABX Subcommittee Approved: March 2017

Formulas for dosing weights: Ideal body weight IBW (male) = 50kg + (2.3 x height in inches > 60 inches) · Ideal body weight IBW (female) = 45kg + (2.3 x height in inches > 60 inches) · Adjusted Body Weight ABW (kg) = IBW + 0.4 (TBW – IBW)

Drug	CrCl > 50 mL/min	CrCl 10 – 50 mL/min	CrCl < 10 mL/min	Intermittent Hemodialysis (IHD)	CRRT
Acyclovir (IV)[1–6] **(Use adjusted BW for obese)**	HSV: 5 mg/kg q8h HSV encephalitis/zoster: 10 mg/kg q8h	Same dose CrCl 25 – 50: q12h CrCl 10 – 25: q24h	HSV: 2.5 mg q24h HSV encephalitis/zoster: 5 mg/kg q24h	HSV: 2.5 mg/kg q24h HSV encephalitis/zoster: 5 mg/kg q24h *Dose daily, but after HD on HD days*	CVVH: 5 – 10 mg/kg q24h CVVHDF: 5 – 10 mg/kg q12h HSV encephalitis/zoster: 10 mg/kg q12h
Acyclovir (PO)[1,2]	HSV mucocutaneous: CrCl > 25: 400 mg q8h, Alt: 200 mg 5x daily; CrCl 10 – 25: 200 mg q8h; CrCl < 10: 200 mg q12h VZV: CrCl > 25: 800 mg q4h (or 5x daily); CrCl 10 – 25: 800 mg q8h; CrCl < 10: 800 mg q12h			See CrCl < 10 mL/min	No data
Amikacin[1,2,5,7,8] **(Use adjusted BW in obese)** **See appendix for complete guidelines**	Conventional dosing: CrCl > 60: 5 – 7.5 mg/kg q8h; CrCl 40 – 60: 5 – 7.5 mg/kg q12h; CrCl 20 – 40: 5 – 7.5 mg/kg q24h; CrCl < 20: 5 mg/kg load, then by level High-dose extended-interval dosing: CrCl > 60: 15 – 20 mg/kg q24h; CrCl 40 – 60: 15 mg/kg q36h; CrCl 20 – 40: CrCl > 30: 15 mg/kg q48h, CrCl < 30: Not recommended; CrCl < 20: alt: 7.5 mg/kg q48–72h **Timing of levels**: Draw trough 30 min prior to 4th dose. Draw peak 30 min after infusion ends **Once daily dosing**: goal peak 35 – 60 mcg/mL; goal trough < 4 mcg/mL **Conventional dosing**: goal peak 25 – 35 mcg/mL for serious infections; 15 – 20 mcg/mL for UTI; goal trough < 4 – 8 mcg/mL			5 – 7.5 mg/kg post HD only consult pharmacist	10 mg/kg load, then 7.5 mg/kg q24–48h Severe/MDR organism: 25 mg/kg q48h consult pharmacist
Amoxicillin (PO)[1,2]	Usual dose: 250 – 500 mg q8h or 875 mg q12h *H pylori*: 1,000 mg q12h Procedural ppx: 2,000 mg x 1	CrCl 10–30: 250 – 500 mg q12h	250 – 500 mg q24h	250 – 500 mg q24h; administer additional dose at the end of dialysis	No data
Amoxicillin/clavulanate (PO)[1,2]	Usual dose: 250 – 500 mg q8h or 875 mg q12h CAP: 2,000 mg ER q12h	CrCl <30: Do not use 875 mg tablet or ER tab CrCl 10 – 30: 250 – 500 mg q12h	250 – 500 mg q24h	250 – 500 mg q24h; administer additional dose at the end of dialysis	No data
Amphotericin B Liposomal[1,2] **(Consider adjusted BW in obese)**	3 – 6 mg/kg/day	No change	No change	No change	No change
Ampicillin (IV)[1–3]	Mild/uncomplicated: 1 – 2 g q6h Meningitis/endovascular/PJI: 2 g q4h	Mild/uncomplicated: 1 g q6–8h Meningitis/endovascular/PJI: 2 g q6–12h	Mild/uncomplicated: 1 g q12h Meningitis/endovascular/PJI: 2 g q12–24h; or 1 g q8h	Mild/uncomplicated: 1 g q12h Meningitis/endovascular/PJI: 2 g q12–24h	CVVH: 2 g q8–12H CVVHDF: 2 g q6–8h Meningitis/endovascular/PJI: 2 g q6h
Ampicillin/sulbactam[1–3,5]	1.5 – 3 g q6h	CrCl < 30: 1.5 – 3 g q12h	CrCl < 15: 1.5 – 3 g q24h	1.5 – 3 g q12–24h *Dose daily, but after HD on HD days*	3 g q6–8h
Azithromycin (IV/PO)[1,2]	500 mg q24h	No change	No change	No change	No change
Aztreonam[1–3,9] Severe: pseudomonas, life-threatening infections	1 – 2 g q8h Severe/Meningitis: 2 g q6–8h	CrCl < 30: 1 g q8h Severe/Meningitis: 1 g q6–8h	500 mg q8h Severe/Meningitis: 1g q12h	1 g q24h Severe/Meningitis: 1 g q12h	2 g load, then 1 g q8h – or – 2 g q12h
Caspofungin[1,2]	70 mg x 1, then 50 mg q24h Optional: 70 mg x 1, then 35 – 50 mg q24h if severe hepatic dysfunction/ESLD. 70 mg q24h if on phenytoin, rifampin, other strong enzyme inducers			No change	No change
Cefazolin[1–5,10]	CrCl ≥ 35: Mild/moderate: 1 g q8h Severe: 2 g q8h	CrCl 10 – 34: Mild/moderate: 0.5 g q12h Severe: 1 g q12h	1 g q24h	1 g q24h *Dose daily, but after HD on HD days* alt: 2g/2g/3g post-HD only	2 g q12h
Cefepime[1–3,5,11,12]	Extended Infusion (4-hour infusion) General: CrCl > 60: 1 g q8h or 2 g q12h; CrCl 30 – 60: 1 g q12h or 2 g q24h; CrCl < 30: 1 g q24h CNS/FN: CrCl > 60: 2 g q8h; CrCl 30 – 60: 2 g q12h; CrCl < 30: 1 g q12h		General: 0.5 g q24h CNS/FN: 1 g q24h	0.5 – 1 g q24h *Dose daily, but after HD on HD days* alt: 2 g post-HD only	2 g load, then 1 g q8h – or – 2 g q12h
Ceftaroline[1,2,13] **(SHC Restriction)**	General: CrCl > 50: 600 mg q12h; CrCl 30 – 50: 400 mg q12h; CrCl 15 – 30: 300 mg q12h; CrCl < 15: 200 mg q12h Endocarditis/ *S.aureus* bacteremia: CrCl > 50: 600 mg q8–12h; CrCl 30 – 50: 400 mg q8–12h; CrCl 15 – 30: 300 mg q8–12h; CrCl < 15: 200 mg q8–12h			200 mg q8–12h Endocarditis/*S.aureus* bacteremia: 200 mg q8–12h	No data
Ceftazidime (IV)[1–3]	Usual dose: 1 – 2 g q8h Severe: 2 g q8h	CrCl 30 – 50: 1 – 2 g q12h CrCl 16 – 30: 1 – 2 g q24h CrCl 6 – 15: 0.5 – 1 g q24h	CrCl < 5: 0.5 g q24h	0.5 – 1 g q24h *Dose daily, but after HD on HD days* alt: 1 – 2 g q48–72h or post-HD only	2 g load, then 1 g q8h – or – 2 g q12h
Ceftazidime/avibactam[1,2,14] **(SHC Restriction)**	2.5 g q8h	CrCl 31 – 50: 1.25 g q8h CrCl 16 – 30: 0.94 g q12h CrCl 6 – 15: 0.94 g q24h	CrCl < 5: 0.94 g q48h	0.94 g q24–48h *Dose daily, but after HD on HD days*	1.25 g q8h

<table>
<tr><th>Drug</th><th>CrCl > 50 mL/min</th><th>CrCl 10 – 50 mL/min</th><th>CrCl < 10 mL/min</th><th>Intermittent Hemodialysis (IHD)</th><th>CRRT</th></tr>
<tr><td>Ceftolozane/tazobactam [1,2,15,16]
(SHC Restriction)</td><td colspan="3"><table>
<tr><td></td><td><u>CrCl > 50</u></td><td><u>CrCl 30 – 50</u></td><td><u>CrCl 15 – 29</u></td><td><u>CrCl < 15</u></td></tr>
<tr><td>General/
CF exacerbation</td><td>1.5 g q8h</td><td>750 mg q8h</td><td>375 mg q8h</td><td rowspan="2">No data</td></tr>
<tr><td>Ventilator-associated pneumonia</td><td>3 g q8h</td><td>1.5 g q8h</td><td>750 mg q8h</td></tr>
</table></td><td>750 mg load, then 150 mg q8h
Administer dose immediately after dialysis on dialysis days</td><td>1.5 g IV q8h</td></tr>
<tr><td>Ceftriaxone (IV) [1,2,17]</td><td colspan="2">1 – 2 g q24h
<u>Endovascular/osteomyelitis/PJI</u>: 2 g q24h
<u>Meningitis, E. faecalis endocarditis</u>: 2 g q12h</td><td>No change</td><td>No change</td><td>No change</td></tr>
<tr><td>Cephalexin (PO) [1,2,18]</td><td>250 – 1000 mg q6h
<u>Uncomplicated cystitis</u>:
500 mg q12h
<u>Cellulitis/SSTI</u>: 500 mg q6h</td><td colspan="2"><u>CrCl 15 – 29</u>: 250 mg q8–12h
<u>CrCl 5 – 14</u>: 250 mg q24h</td><td>500 mg q24h
Dose daily, but after HD on HD days</td><td>No data</td></tr>
<tr><td>Cefpodoxime (PO) [1,2]</td><td><u>Uncomplicated cystitis</u>:
100 mg q12h
<u>CAP/bronchitis</u>: 200 mg q12h
<u>Skin/skin structure</u>:
400 mg q12h</td><td colspan="2"><u>CrCl < 30</u>: same dose q24h</td><td>Same dose, post-HD only</td><td>No data</td></tr>
<tr><td>Ciprofloxacin (IV/PO) [1–4,19]</td><td colspan="3"><table>
<tr><td></td><td><u>CrCl > 50</u></td><td><u>CrCl 30 – 50</u></td><td><u>CrCl < 30</u></td></tr>
<tr><td>General infections</td><td>400 mg IV q12h
500 mg PO q12h</td><td>Same</td><td>400 mg IV q24h
500 mg PO q24h</td></tr>
<tr><td>Pseudomonas, severe</td><td>400 mg IV q8h
750 mg PO q12h</td><td>400 mg IV q8–12h
500 mg PO q12h</td><td>400 mg IV q24h
500 mg PO q24h</td></tr>
</table></td><td>200 – 400 mg IV q24h
250 – 500 mg PO q24h
Dose daily, but after HD on HD days</td><td>400 mg IV q12–24h
500 mg PO q12–24h

<u>Septic pt > 90 kg on CVVHF/CVVHDF with A.baumannii or P.aeruginosa</u>:
400 mg IV q12–8h</td></tr>
<tr><td>Clindamycin [1,2]</td><td>600 – 900 mg IV q8h
150 – 450 mg PO q6h</td><td>No change</td><td>No change</td><td>No change</td><td>No change</td></tr>
<tr><td>Colistin (IV) [1–3,20–22]
(SHC Restriction)
(Dosage expressed in terms of colistin base activity [CBA]; Use ideal BW in obese)</td><td colspan="3">U.S. FDA Package Insert<table>
<tr><td></td><td><u>CrCl > 80</u></td><td><u>CrCl 50 – 79</u></td><td><u>CrCl 30 – 49</u></td><td><u>CrCl < 30</u></td></tr>
<tr><td>Loading Dose</td><td colspan="4">5 mg/kg x 1 (max dose: 300 mg)</td></tr>
<tr><td>Maintenance Dose</td><td>1.25 – 2.5 mg/kg q12h</td><td>1.25 – 1.9 mg/kg q12h</td><td>2.5 mg/kg q24h</td><td>1.5 mg/kg q36h</td></tr>
</table>
Preferred Dosing for Critically Ill Patients (Consult ID Pharmacist)<table>
<tr><td></td><td>CrCl</td><td>Dosing Regimen</td></tr>
<tr><td>Loading Dose</td><td>All CrCl
(including HD and CRRT)</td><td>= 4 x IBW (kg)
(max dose: 300 mg)</td></tr>
<tr><td rowspan="11">Maintenance Dose</td><td>> 90 mL/min</td><td>180 mg q12h</td></tr>
<tr><td>80 – 89 mL/min</td><td>170 mg q12h</td></tr>
<tr><td>70 – 79 mL/min</td><td>150 mg q12h</td></tr>
<tr><td>60 – 69 mL/min</td><td>138 mg q12h</td></tr>
<tr><td>50 – 59 mL/min</td><td>122 mg q12h</td></tr>
<tr><td>40 – 49 mL/min</td><td>110 mg q12h</td></tr>
<tr><td>30 – 39 mL/min</td><td>98 mg q12h</td></tr>
<tr><td>20 – 29 mL/min</td><td>88 mg q12h</td></tr>
<tr><td>10 – 19 mL/min</td><td>80 mg q12h</td></tr>
<tr><td>5 – 9 mL/min</td><td>72 mg q12h</td></tr>
<tr><td>0 mL/min</td><td>65 mg q12h</td></tr>
</table>
Suggested loading dose and daily doses of colistimethate for desired target colistin Css,avg of 2 mg/L (CID 2017:64. Nation et al)</td><td><u>Loading Dose</u>: 4 x IBW (kg)
(max dose: 300 mg)

<u>Maintenance Dose</u>:
65 mg q12h; then supplement an additional 40 mg (for a 3-hr IHD session) or 50 mg (for a 4-hr IHD session) post-dialysis

<u>alt</u>: 1.5 mg/kg q24h</td><td><u>Loading Dose</u>: 4 x IBW (kg)
(max dose: 300 mg)

<u>Maintenance Dose</u>:
220 mg q12h

<u>alt</u>: 2.5 mg/kg q24h</td></tr>
<tr><td>Daptomycin [1,2,23–29]
(SHC Restriction)
(Use adjusted BW in obese)</td><td><u>Skin/Soft tissue</u>:
6 mg/kg q24h
<u>Bacteremia/Endovascular</u>:
8 mg/kg q24h
(If VRE, doses up to 10 – 12 mg/kg q24h; consult ID)</td><td><u>CrCl < 30</u>:
Same dose q48h</td><td>Same dose q48h</td><td>Same dose q48h
Dose q48h, but after HD on HD days

<u>alt</u>: ≥ 6 mg/kg post-HD only or 6/6/9 mg/kg post-HD only</td><td>6 – 10 mg/kg q48h

<u>alt</u>: 4 – 8 mg/kg q24h</td></tr>
<tr><td>Doxycycline (IV/PO) [1,2]</td><td>100 mg q12h</td><td>No change</td><td>No change</td><td>No change</td><td>No change</td></tr>
<tr><td>Ertapenem (IV/IM) [1,2]</td><td>1 g q24h</td><td><u>CrCl <30</u>: 500 mg q24h</td><td>500 mg q24h</td><td>500 mg q24h
Dose daily, but after HD on HD days</td><td>1 g q24h</td></tr>
<tr><td>Ethambutol (PO) [1,5,30,31]
(Use lean BW if obese)
(See footnote for lean BW equation)</td><td><u>Dose range</u>:
15 – 25 mg/kg/day
(max dose: 1,600 mg/day)<table>
<tr><td>Lean body weight</td><td>Dose</td></tr>
<tr><td>40 – 55 kg</td><td>800 mg</td></tr>
<tr><td>56 – 75 kg</td><td>1,200 mg</td></tr>
<tr><td>76 – 90 kg</td><td>1,600 mg</td></tr>
</table></td><td><u>CrCl 10 – 50</u>:
15 – 25 mg/kg q24–36h</td><td><u>CrCl < 10</u>:
15 – 25 mg/kg q48h</td><td>15 – 25 mg/kg 3 times per week post-HD
Administer after HD only</td><td>15 – 25 mg/kg q24–36h</td></tr>
<tr><td>Fidaxomicin (PO) [1,2]
(SHC Restriction)</td><td>200 mg q12h x 10 days</td><td>No change</td><td>No change</td><td>No change</td><td>No change</td></tr>
<tr><td>Fluconazole (IV/PO) [1–4,32]
Dose by indication. Load 800 mg for candidemia</td><td>200 – 400 mg q24h
<u>C.glabrata candidemia</u>:
800 mg q24h
<u>Severe/CNS infections</u>:
up to 800 mg q24h</td><td colspan="2">(50% of normal dose) q24h</td><td>Dose by indication;
200 – 800 mg post HD only</td><td>400 – 800 mg q24h</td></tr>
</table>

Drug	CrCl > 50 mL/min	CrCl 10 – 50 mL/min	CrCl < 10 mL/min	Intermittent Hemodialysis (IHD)	CRRT

Foscarnet (IV)[1,2,33–35]
(Consider adjusted BW in obese)

Adj CrCl (mL/min/kg)
$\left(\frac{140 - age}{SCr \times 72}\right) X\ (0.85 \text{ if female})$

CrCl (mL/min/kg)	CMV *induction*		CMV *maintenance*		HSV	
> 1.4	60 mg/kg q8h	90 mg/kg q12h	90 mg/kg q24h	120 mg/kg q24h	40 mg/kg q12h	40 mg/kg q8h
> 1.0 – 1.4	45 mg/kg q8h	70 mg/kg q12h	70 mg/kg q24h	90 mg/kg q24h	30 mg/kg q12h	30 mg/kg q8h
> 0.8 – 1.0	50 mg/kg q12h	50 mg/kg q12h	50 mg/kg q24h	65 mg/kg q24h	20 mg/kg q12h	35 mg/kg q12h
> 0.6 – 0.8	40 mg/kg q12h	80 mg/kg q24h	80 mg/kg q48h	105 mg/kg q48h	35 mg/kg q24h	25 mg/kg q12h
> 0.5 – 0.6	60 mg/kg q24h	60 mg/kg q24h	60 mg/kg q48h	80 mg/kg q48h	25 mg/kg q24h	40 mg/kg q24h
≥ 0.4 – 0.5	50 mg/kg q24h	50 mg/kg q24h	50 mg/kg q48h	65 mg/kg q48h	20 mg/kg q24h	35 mg/kg q24h
< 0.4	Not recommended		Not recommended		Not recommended	
IHD	60 – 90 mg/kg loading dose, then 45 – 60 mg/kg/dose post-HD only		No data		No data	
CRRT	No data					

Ganciclovir (IV)[1,2]
(Consider adjusted BW in obese)

CMV	CrCl >70*	CrCl >50	CrCl >25	CrCl >10	CrCl <10
Induction (I)	5 mg/kg q12h	2.5 mg/kg q12h	2.5 mg/kg q24h	1.25 mg/kg q24h	1.25 mg/kg 3x/week
Maintenance (M)	5 mg/kg q24h	2.5 mg/kg q24h	1.25 mg/kg q24h	0.625 mg/kg q24h	0.625 mg/kg 3x/week

*Manufacturer's CrCl cutoffs. **Please refer to BMT protocols if applicable**

IHD: I: 1.25 mg/kg post HD only; M: 0.625 mg/kg post HD only

CRRT: I: 2.5 mg/kg q12–24h; M: 1.25 – 2.5 mg/kg q24h

Gentamicin[1,3,36]
(Use adjusted BW in obese)

See appendix for complete guidelines

	CrCl > 60	CrCl 40 – 59	CrCl 20 – 39	CrCl < 20	IHD	CRRT
Gram negative	1.7 mg/kg q8h ***or*** 5 – 7 mg/kg q24h *(high-dose extended-interval)*	1.7 mg/kg q12h ***or*** 5 – 7 mg/kg q36h *(high-dose extended-interval)*	1.7 mg/kg q24h ***or*** CrCl > 30: 5 – 7 mg/kg q48h CrCl < 30: Not recommended *(high-dose extended-interval)*	2 mg/kg loading dose, then per level	2 mg/kg loading dose, then 1.5 mg/kg post HD	1.5 – 2.5 mg/kg q24–48h
Gram positive synergy	1 mg/kg q8h**	1 mg/kg q12h	1 mg/kg q24h	1 mg/kg load, then by level	1 mg/kg q48–72h; consider redosing when level < 1 mcg/L	1 mg/kg q24h, then per level

Goal levels: **Gram-negative infections:** Goal peak for traditional dosing 4 – 8 mcg/mL; goal trough < 1 – 2 mcg/mL
Gram-positive synergy: Goal peak 3 – 4 mcg/mL; goal trough < 1 mcg/mL

Timing of levels: Draw peak 30 minutes after completion of 3rd dose. Draw trough 30 minutes prior to 4th dose (For CrCl < 20 mL/min, may check levels sooner than 3rd/4th dose)
For 7 mg/kg once-daily dosing, draw a single random level 8 – 12 hours after dose administration. Adjust based on **Hartford nomogram**
For HD, draw trough pre-HD (alternative: draw trough level 4-hr post-HD); and peak 30 minutes after end of each infusion

** Streptococci, *Streptococcus gallolyticus (bovis)*, *Streptococcus viridans* endocarditis: optional dosing 3 mg/kg q24h for CrCl > 60 mL/min
** Staphylococci; Enterococcus spp (strains susceptible to PCN and gentamicin) endocarditis: optional dosing 3 mg/kg in 2 or 3 equally divided doses

Drug	CrCl > 50 mL/min	CrCl 10 – 50 mL/min	CrCl < 10 mL/min	Intermittent Hemodialysis (IHD)	CRRT
Isavuconazole (IV/PO)[1,2] **(SHC Restriction)**	Initial: 372 mg q8h x 6 doses Maintenance: 372 mg q24h	No change	No change	No change	No change
Isoniazid (PO)[1,2,30,31]	300 mg q24h (5 mg/kg/day)	No change	No change	No change	No change

Levofloxacin (IV/PO)[1–4]

	CrCl ≥ 50	CrCl 20 – 49	CrCl < 20
General	250 – 500 mg q24h	250 mg q24h - or - 500 mg q48h	500 mg x1, then 250 mg q48h
Severe/PNA/ Pseudomonas/ Stenotrophomonas:	750 mg q24h	750 mg q48h	750 mg x1, then 500 mg q48h

IHD: See CrCl < 20 ml/min
Dose q48h, but after HD on HD days

CRRT: 750 mg load, then 250 – 750 mg q24h

Drug	CrCl > 50 mL/min	CrCl 10 – 50 mL/min	CrCl < 10 mL/min	Intermittent Hemodialysis (IHD)	CRRT
Linezolid (IV/PO)[1,2] **(SHC Restriction)**	600 mg q12h	No change	No change	No change	No change

Meropenem[1–4,37]
(SHC Restriction)
3-hr extended infusion

	CrCl > 50	CrCl 26 – 50	CrCl 10 – 25	CrCl < 10
Usual dose (FN, PNA, Pseudomonas)	0.5 – 1 g q8h	0.5 – 1 g q12h	0.5 g q12h	0.5 g q24h
CF/Meningitis	2 g q8h	2 g q12h	1 g q12h	1 g q24h

IHD: 500 mg q24h
CF/CNS: 1 g q24h
Dose daily, but after HD on HD days

CRRT: 1 g q8–12h – or – 500 mg q6–8h
CF/CNS: 2 g q12h

Drug	CrCl > 50 mL/min	CrCl 10 – 50 mL/min	CrCl < 10 mL/min	Intermittent Hemodialysis (IHD)	CRRT
Metronidazole (IV/PO)[1,2]	500 mg q6–8h	No change Severe hepatic impairment: can consider 500 mg q12h		500 mg q8h	500 mg q6–8h
Moxifloxacin (IV/PO)[1,2]	400 mg IV/PO q24h	No change	No change	No change	No change
Nafcillin[1,2]	2 g q4h Mild infections: 1 g q4h	No change for renal impairment. Hepatic Impairment: No specific dose adjustment provided by manufacturer. Dosage adjustment may be necessary in the setting of concomitant renal impairment; nafcillin primarily undergoes hepatic metabolism.			

Oseltamivir (PO)[1,2,38]

	CrCl ≥ 60	CrCl 30 – 60	CrCl 10 – 30
Prophylaxis	75 mg q24h	30 mg q24h	30 mg q48h
Treatment	75 mg q12h	30 mg q12h	30 mg q24h

IHD: Prophylaxis: 30 mg x 1, then 30 mg after every other HD session
Treatment: 30 mg x 1, then 30 mg post-HD only

CRRT: Prophylaxis: 75 mg q24h
Treatment: 75 mg q12h

Drug	CrCl > 50 mL/min	CrCl 10 – 50 mL/min	CrCl < 10 mL/min	Intermittent Hemodialysis (IHD)	CRRT
Penicillin G (IV)[1–3,5]	2 – 4 mu q4h Dose range: 12 – 24 million units/day continuous infusion or in divided doses every 4 to 6 hours	2 – 3 mu q4h	1 – 2 mu q6h	Mild: 0.5 – 1 mu q4–6h; or 1 – 2 mu q8–12h Severe: 2 mu q4–6h; or 4 mu q8–12h	4 mu q4–6h

<table>
<tr><th>Drug</th><th>CrCl > 50 mL/min</th><th>CrCl 10 – 50 mL/min</th><th>CrCl < 10 mL/min</th><th>Intermittent Hemodialysis (IHD)</th><th>CRRT</th></tr>
<tr><td>Piperacillin/tazobactam[1–4,39,40]</td><td colspan="3">
<table>
<tr><th></th><th><u>CrCl > 40</u></th><th><u>CrCl 20 – 40</u></th><th><u>CrCl < 20</u></th></tr>
<tr><td colspan="4">Intermittent Dosing</td></tr>
<tr><td>General</td><td>3.375 g q6h</td><td>2.25 g q6h</td><td>2.25 g q8h</td></tr>
<tr><td>Severe/sepsis/CF/ nosocomial PNA</td><td>4.5 g q6h</td><td>3.375 g q6h</td><td>2.25 g q6h</td></tr>
<tr><td colspan="4">Extended-Infusion Dosing (4-hr infusion)</td></tr>
<tr><td>General, CF Pseudomonas, nosocomial PNA:</td><td colspan="2"><u>Extended infusion for CrCl > 20</u>:
3.375 – 4.5 g q8h over 4h*</td><td>3.375 g q12h over 4h</td></tr>
</table>
*In select cases, higher piperacillin/tazobactam dosing may be warranted, e.g. sepsis, critically ill patients with severe or deep seated infections, infections with MIC > 16 mg/L, obesity with weight > 120kg or BMI > 40, CrCl > 120 mL/min, or enhanced drug clearance such as those with cystic fibrosis: consider doses of 4.5 g q8h (infused over 4 hours) or q6h.</td>
<td><u>General</u>: 2.25 g q12h

<u>Severe infections</u>:
3.375 g q12h over 4-hr

<u>alt</u>: 2.25 g q8h</td>
<td>3.375 g q6h

<u>Extended infusion</u>:
3.375 – 4.5 g q8h over 4-hr</td></tr>
<tr><td>Polymyxin B[1,2,41]
(SHC Restriction)
(Use adjusted BW if obese)</td><td colspan="3">7,500 – 12,500 units/kg q12h
(maximum: 25,000 units/kg/day)</td><td>No data</td><td>No change</td></tr>
<tr><td>Posaconazole (PO/IV)[1,2]
(SHC Restriction [IV])</td><td colspan="3">
<table>
<tr><th></th><th>Oral Suspension</th><th>Delayed-release tablet / Intravenous solution</th></tr>
<tr><td>Prophylaxis</td><td>200 mg q8h</td><td rowspan="2">300 mg q12h x 1 day, then 300 mg q24h</td></tr>
<tr><td>Treatment</td><td><u>Usual dose</u>: 200 mg q6–8h or 400 mg q12h</td></tr>
</table>
No renal adjustment
Delayed-release tablet and oral suspension are not interchangeable
Posaconazole levels shown to have great degree of interpatient variability. Consider drawing a trough 4 – 7 days after initiating dose</td>
<td>No change</td><td>No change</td></tr>
<tr><td>Pyrazinamide (PO)[1,2,30,31]
(Use lean BW if obese)
(See footnote for lean BW equation)</td><td><u>Usual Dose</u>: 25 mg/kg q24h
(max dose: 2,000 mg/day)
<table>
<tr><th>Lean body weight</th><th>Dose</th></tr>
<tr><td>40 – 55 kg</td><td>1,000 mg</td></tr>
<tr><td>56 – 75 kg</td><td>1,500 mg</td></tr>
<tr><td>76 – 90 kg</td><td>2,000 mg</td></tr>
</table></td>
<td colspan="2"><u>CrCl < 30</u>:
25 mg/kg 3 times per week</td><td>25 mg/kg 3 times per week
Administer after HD only</td><td>No data</td></tr>
<tr><td>Rifampin (IV/PO)[1,2,30,31,42–44]
Capsule size: 150mg, 300mg</td><td colspan="2"><u>TB</u>: 600 mg q24h (≤ 45 kg: 10 mg/kg q24h)
<u>Endocarditis</u>: 300 mg q8h
<u>PJI</u>: 300 – 450 mg q12h
<u>Vertebral Osteomyelitis</u>: 600 mg q24h</td><td>No change</td><td>No change</td><td>No change</td></tr>
<tr><td>Tedizolid (IV/PO)[1,2,45]
(SHC Restriction)</td><td>200 mg q24h</td><td>No change</td><td>No change</td><td>No change</td><td>No change</td></tr>
<tr><td>Tobramycin[1,2,36]</td><td colspan="5">Refer to Gentamicin for dosing. See appendix for complete guidelines.</td></tr>
<tr><td>Trimethoprim (TMP)/ Sulfamethoxazole (IV/PO)[1,2,4,46]

(Dose by adjusted BW in obese)
SS = 80 mg TMP = 10 ml po soln
DS =160 mg TMP = 20ml po soln</td><td><u>Usual Dose Range</u>:
PO: 1 – 2 DS tabs q12–24h
IV: 8 – 20 mg/kg/day TMP divided q6–12h
<u>UTI</u>: 1 DS tab PO BID
<u>SSTI</u>: 1 – 2 DS tab PO BID
<u>PCP/Stenotrophomonas</u>:
15 – 20 mg/kg/day TMP divided q6–8h (approximately 2 DS tab q8h)</td><td><u>CrCl 15 – 30</u>:
Administer 50% of recommended dose

<u>PCP/Stenotrophomonas</u>:
7.5 – 10 mg/kg/day TMP divided q8–12h</td><td><u>CrCl < 15</u>:
Use is not recommended, but if needed for PCP/Stenotrophomonas:
5 – 10 mg/kg TMP q24h</td><td>2.5 – 5 mg/kg TMP q24h

<u>PCP/Stenotrophomonas</u>:
5 – 10 mg/kg TMP q24h
Dose daily, but after HD on HD days

<u>alt</u>: 5 – 20 mg/kg TMP post-HD only</td><td>5 – 10 mg/kg/day TMP divided q12h

<u>PCP/ Stenotrophomonas</u>:
15 mg/kg/day TMP divided q8–12h</td></tr>
<tr><td>Valacyclovir (PO)[1,2]

Please refer to transplant protocols if applicable</td><td colspan="3">
<table>
<tr><th></th><th><u>CrCl > 30</u></th><th><u>CrCl 10 – 30</u></th><th><u>< 10</u></th></tr>
<tr><td>VZV</td><td><u>CrCl >50</u>: 1 g q8h
<u>CrCl 30-50</u>: 1 g q12h</td><td>1 g q24h</td><td>500 mg q24h</td></tr>
<tr><td>Genital herpes</td><td><u>Initial episode</u>: 1 g q12h
<u>Recurrent episode</u>:
500 mg q12h</td><td><u>Initial episode</u>: 1 g q24h
<u>Recurrent</u>: 500 mg q24h</td><td><u>Initial/recurrent episode</u>:
500 mg q24h</td></tr>
<tr><td>Herpes labialis</td><td><u>CrCl >50</u>:
2 g q12h x 2 doses
<u>CrCl 30 – 50</u>:
1 g q12h x 2 doses</td><td>500 mg q12h x 2 doses</td><td>500 mg x 1 dose</td></tr>
</table></td>
<td>500 mg q24h
Dose daily, but after HD on HD days</td><td>No data</td></tr>
<tr><td>Valganciclovir (PO)[1,2]
Please refer to transplant protocols if applicable</td><td colspan="5">
<table>
<tr><th></th><th><u>CrCl > 60</u></th><th><u>CrCl 40 – 59</u></th><th><u>CrCl 25 – 39</u></th><th><u>CrCl 10 – 24</u></th><th><u>CrCl < 10; IHD</u></th><th><u>CRRT</u></th></tr>
<tr><td>Induction (14-21 days)</td><td>900 mg q12h</td><td>450 mg q12h</td><td>450 mg q24h</td><td>450 mg q48h</td><td>200 mg 3x/week after HD only</td><td>No data</td></tr>
<tr><td>Maintenance/ prophylaxis</td><td>900 mg q24h</td><td>450 mg q24h</td><td>450 mg q48h</td><td>450 mg twice/week</td><td>100 mg 3x/week after HD only</td><td>No data</td></tr>
</table></td></tr>
<tr><td>Vancomycin (IV)[1,2,47,48]

(Use actual body weight; refer to Vancomycin Guide Appendix C for obesity dosing)</td><td colspan="3">Consider loading dose of 25 – 30 mg/kg (max 2.5 g) for severe infections and ICU
<table>
<tr><th>CrCl (mL/min)</th><th>Dose & Frequency</th><th>Total daily dose range</th></tr>
<tr><td>> 90</td><td>15 mg/kg q8h to 15 – 20 mg/kg q12h</td><td>30 – 45 mg/kg/day</td></tr>
<tr><td>51 – 89</td><td>15 – 20 mg/kg q12h</td><td>30 – 40 mg/kg/day</td></tr>
<tr><td>30 – 50</td><td>15 mg/kg q12h to 20 mg/kg q24h</td><td>20 – 30 mg/kg/day</td></tr>
<tr><td>10 – 29</td><td>10 – 15 mg/kg q24h to 15 mg/kg q48h</td><td>7.5 – 15 mg/kg/day</td></tr>
<tr><td>< 10 or AKI</td><td>15 mg/kg x 1, then dose by level</td><td>N/A</td></tr>
</table>
Goal trough 10 – 15 mcg/mL (cellulitis, skin/soft tissue infections)
Goal trough 15 – 20 mcg/mL (pneumonia, S. Aureus bacteremia, endocarditis, osteomyelitis)
Timing of levels: Draw trough < 30 minutes before 4th dose of new regimen. When SCr acutely rises, hold dose, restart when level < 15 – 20 mcg/mL
See appendix for complete guidelines</td>
<td>15 – 20 mg/kg x 1, then redose per algorithm

(see Appendix E of Vancomycin per Pharmacy Protocol)</td><td>15 – 20 mg/kg x 1, then 10 – 15 mg/kg q24h

Draw level prior to 3rd dose. Adjust to levels</td></tr>
<tr><td>Vancomycin PO[1,2,49]</td><td colspan="3">Poor systemic absorption- used for the treatment of Clostridium difficile-associated diarrhea
<u>Mild/moderate/severe</u>: 125 mg PO q6h
<u>Severe complicated (CDI-related septic shock, ileus, toxic megacolon)</u>: 500 mg PO q6h</td><td>No change</td><td>No change</td></tr>
</table>

Drug	CrCl > 50 mL/min	CrCl 10 – 50 mL/min	CrCl < 10 mL/min	Intermittent Hemodialysis (IHD)	CRRT
Voriconazole (IV/PO)[1,2,50,51] **(Dose by adjusted BW in obese)**	IV: 6 mg/kg IV q12h x 2, then 4 mg/kg IV q12h PO: 400 mg PO q12h x 2, then 200 mg PO q12h	**IV→PO conversion 1:1 (round to nearest tablet size- available in 200 mg and 50 mg tablets)** Caution with IV: accumulation of IV vehicle cyclodextran occurs. Consider PO if CrCl < 50 mL/min unless benefits justify risks of IV use. Levels shown to have great degree of interpatient variability. Consider drawing a trough 4 – 7 days after new dose.			

Abbreviations: SCr = serum creatinine; LD = loading dose; MU= million units; PNA = pneumonia; HD = hemodialysis; CAP = community acquired pneumonia; CRRT = continuous renal replacement therapy; TMP = trimethoprim; PCP: pneumocystis jiroveci pneumonia; TB = tuberculosis; UF = ultrafiltration

CRRT dosing: doses listed are for CVVHDF and CVVHD modalities, which are the most common modes at SHC. Note that these are generally higher than doses used in CVVH.

LBW (men) = (1.10 x Weight(kg)) - 128 x (Weight^2/(100 x Height(m))^2)

LBW (women) = (1.07 x Weight(kg)) - 148 x (Weight^2/(100 x Height(m))^2)

LBW online calculator: http://www.empr.com/medical-calculators/lean-body-weight-calculator/article/170219/

References:

1. Lexicomp Online. http://online.lexi.com. Accessed April 9, 2017.
2. MICROMEDEX®. http://www.micromedexsolutions.com.laneproxy.stanford.edu/micromedex2/librarian. Accessed April 10, 2017.
3. Heintz BH, Matzke GR, Dager WE. Antimicrobial Dosing Concepts and Recommendations for Critically Ill Adult Patients Receiving Continuous Renal Replacement Therapy or Intermittent Hemodialysis. *Pharmacother J Hum Pharmacol Drug Ther*. 2009;29. doi:10.1592/phco.29.5.562.
4. Trotman RL, Williamson JC, Shoemaker DM, Salzer WL. Antibiotic Dosing in Critically Ill Adult Patients Receiving Continuous Renal Replacement Therapy. *Clin Infect Dis*. 2005;41(8):1159-1166. doi:10.1086/444500.
5. Aronoff G, Bennett W, Berns J, et al. *Drug Prescribing in Renal Failure*. 5th ed. Philadelphia, PA: American College of Physicians; 2007.
6. Turner RB, Cumpston A, Sweet M, et al. Prospective, Controlled Study of Acyclovir Pharmacokinetics in Obese Patients. *Antimicrob Agents Chemother*. 2016;60. doi:10.1128/aac.02010-15.
7. Roger C, Wallis SC, Muller L, et al. Influence of Renal Replacement Modalities on Amikacin Population Pharmacokinetics in Critically Ill Patients on Continuous Renal Replacement Therapy. *Antimicrob Agents Chemother*. 2016;60. doi:10.1128/aac.00828-16.
8. Taccone FS, Backer D de, Laterre P-F, et al. Pharmacokinetics of a loading dose of amikacin in septic patients undergoing continuous renal replacement therapy. *Int J Antimicrob Agents*. 2011;37. doi:10.1016/j.ijantimicag.2011.01.026.
9. Gerig JS, Bolton ND, Swabb EA, Scheld WM, Bolton WK. Effect of hemodialysis and peritoneal dialysis on aztreonam pharmacokinetics. *Kidney Int*. 1984;26. doi:10.1038/ki.1984.174.
10. Stryjewski ME, Szczech LA, Benjamin DK, et al. Use of Vancomycin or First-Generation Cephalosporins for the Treatment of Hemodialysis-Dependent Patients with Methicillin-Susceptible Staphylococcus aureus Bacteremia. *Clin Infect Dis*. 2007;44. doi:10.1086/510386.
11. Crandon JL, Bulik CC, Kuti JL, Nicolau DP. Clinical Pharmacodynamics of Cefepime in Patients Infected with Pseudomonas aeruginosa. *Antimicrob Agents Chemother*. 2010;54. doi:10.1128/AAC.01183-09.
12. Bauer KA, West JE, O'Brien JM, Goff DA. Extended-infusion cefepime reduces mortality in patients with Pseudomonas aeruginosa infections. *Antimicrob Agents Chemother*. 2013;57. doi:10.1128/AAC.02365-12.
13. Vidaillac C, Leonard SN, Rybak MJ. In vitro activity of ceftaroline against methicillin-resistant Staphylococcus aureus and heterogeneous vancomycin-intermediate S. aureus in a hollow fiber model. *Antimicrob Agents Chemother*. 2009;53(11):4712-4717. doi:10.1128/AAC.00636-09.
14. Wenzler E, Bunnell KL, Bleasdale SC, Benken S, Danziger LH, Rodvold KA. Pharmacokinetics and Dialytic Clearance of Ceftazidime-Avibactam in a Critically Ill Patient on Continuous Venovenous Hemofiltration. *Antimicrob Agents Chemother*. 2017;61(7). doi:10.1128/AAC.00464-17.
15. Bremmer DN, Nicolau DP, Burcham P, Chunduri A, Shidham G, Bauer KA. Ceftolozane/Tazobactam Pharmacokinetics in a Critically Ill Adult Receiving Continuous Renal Replacement Therapy. *Pharmacother J Hum Pharmacol Drug Ther*. 2016;36(5):e30-e33. doi:10.1002/phar.1744.
16. Oliver WD, Heil EL, Gonzales JP, et al. Ceftolozane-Tazobactam Pharmacokinetics in a Critically Ill Patient on Continuous Venovenous Hemofiltration. *Antimicrob Agents Chemother*. 2016;60. doi:10.1128/aac.02608-15.
17. Laville M, Mercatello A, Freney J, et al. Pharmacokinetics of ceftriaxone in hemodialysis. *Pathol Biol (Paris)*. 1987;35(5 Pt 2):719-723.
18. Stevens DL, Bisno AL, Chambers HF, et al. Practice Guidelines for the Diagnosis and Management of Skin and Soft Tissue Infections: 2014 Update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2014;59. doi:10.1093/cid/ciu296.
19. Roger C, Wallis SC, Louart B, et al. Comparison of equal doses of continuous venovenous haemofiltration and haemodiafiltration on ciprofloxacin population pharmacokinetics in critically ill patients. *J Antimicrob Chemother*. 2016;71(6):1643-1650. doi:10.1093/jac/dkw043.
20. Nation RL, Garonzik SM, Thamlikitkul V, et al. Dosing Guidance for Intravenous Colistin in Critically Ill Patients. *Clin Infect Dis*. 2017;64(5):565-571. doi:10.1093/cid/ciw839.
21. Plachouras D, Karvanen M, Friberg LE, et al. Population pharmacokinetic analysis of colistin methanesulfonate and colistin after intravenous administration in critically ill patients with infections caused by gram-negative bacteria. *Antimicrob Agents Chemother*. 2009;53(8):3430-3436. doi:10.1128/AAC.01361-08.
22. Dalfino L, Puntillo F, Mosca A, et al. High-dose, extended-interval colistin administration in critically ill patients: is this the right dosing strategy? A preliminary study. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2012;54(12):1720-1726. doi:10.1093/cid/cis286.
23. Dvorchik BH, Damphousse D. The pharmacokinetics of daptomycin in moderately obese, morbidly obese, and matched nonobese subjects. *J Clin Pharmacol*. 2005;45(1):48-56. doi:10.1177/0091270004269562.
24. Pai MP, Norenberg JP, Anderson T, et al. Influence of morbid obesity on the single-dose pharmacokinetics of daptomycin. *Antimicrob Agents Chemother*. 2007;51(8):2741-2747. doi:10.1128/AAC.00059-07.
25. Haselden M, Leach M, Bohm N. Daptomycin dosing strategies in patients receiving thrice-weekly intermittent hemodialysis. *Ann Pharmacother*. 2013;47(10):1342-1347. doi:10.1177/1060028013503110.
26. Patel N, Cardone K, Grabe DW, et al. Use of pharmacokinetic and pharmacodynamic principles to determine optimal administration of daptomycin in patients receiving standardized thrice-weekly hemodialysis. *Antimicrob Agents Chemother*. 2011;55(4):1677-1683. doi:10.1128/AAC.01224-10.
27. Falcone M, Russo A, Cassetta MI, et al. Daptomycin serum levels in critical patients undergoing continuous renal replacement. *J Chemother Florence Italy*. 2012;24(5):253-256. doi:10.1179/1973947812Y.0000000033.
28. Preiswerk B, Rudiger A, Fehr J, Corti N. Experience with daptomycin daily dosing in ICU patients undergoing continuous renal replacement therapy. *Infection*. 2013;41(2):553-557. doi:10.1007/s15010-012-0300-3.
29. Xu X, Khadzhynov D, Peters H, et al. Population pharmacokinetics of daptomycin in adult patients undergoing continuous renal replacement therapy. *Br J Clin Pharmacol*. 2017;83(3):498-509. doi:10.1111/bcp.13131.
30. Drug-Resistant Tuberculosis: A Survival Guide for Clinicians, 3rd edition | Curry International Tuberculosis Center. http://www.currytbcenter.ucsf.edu/products/cover-pages/drug-resistant-tuberculosis-survival-guide-clinicians-3rd-edition. Accessed April 10, 2017.
31. Nahid P, Dorman SE, Alipanah N, et al. Official American Thoracic Society/Centers for Disease Control and Prevention/Infectious Diseases Society of America Clinical Practice Guidelines: Treatment of Drug-Susceptible Tuberculosis. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2016;63(7):e147-195. doi:10.1093/cid/ciw376.
32. Pappas PG, Kauffman CA, Andes DR, et al. Clinical Practice Guideline for the Management of Candidiasis: 2016 Update by the Infectious Diseases Society of America. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2016;62(4):e1-50. doi:10.1093/cid/civ933.
33. Aweeka FT, Jacobson MA, Martin-Munley S, et al. Effect of renal disease and hemodialysis on foscarnet pharmacokinetics and dosing recommendations. *J Acquir Immune Defic Syndr Hum Retrovirology Off Publ Int Retrovirology Assoc*. 1999;20(4):350-357.
34. Jayasekara D, Aweeka FT, Rodriguez R, Kalayjian RC, Humphreys MH, Gambertoglio JG. Antiviral therapy for HIV patients with renal insufficiency. *J Acquir Immune Defic Syndr 1999*. 1999;21(5):384-395.

35. MacGregor RR, Graziani AL, Weiss R, Grunwald JE, Gambertoglio JG. Successful foscarnet therapy for cytomegalovirus retinitis in an AIDS patient undergoing hemodialysis: rationale for empiric dosing and plasma level monitoring. *J Infect Dis*. 1991;164(4):785-787.
36. Nicolau DP, Freeman CD, Belliveau PP, Nightingale CH, Ross JW, Quintiliani R. Experience with a once-daily aminoglycoside program administered to 2,184 adult patients. *Antimicrob Agents Chemother*. 1995;39(3):650-655.
37. Kuti JL, Dandekar PK, Nightingale CH, Nicolau DP. Use of Monte Carlo simulation to design an optimized pharmacodynamic dosing strategy for meropenem. *J Clin Pharmacol*. 2003;43(10):1116-1123. doi:10.1177/0091270003257225.
38. Robson R, Buttimore A, Lynn K, Brewster M, Ward P. The pharmacokinetics and tolerability of oseltamivir suspension in patients on haemodialysis and continuous ambulatory peritoneal dialysis. *Nephrol Dial Transplant Off Publ Eur Dial Transpl Assoc - Eur Ren Assoc*. 2006;21(9):2556-2562. doi:10.1093/ndt/gfl267.
39. Lodise TP, Lomaestro B, Drusano GL. Piperacillin-tazobactam for Pseudomonas aeruginosa infection: clinical implications of an extended-infusion dosing strategy. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2007;44(3):357-363. doi:10.1086/510590.
40. Patel N, Scheetz MH, Drusano GL, Lodise TP. Identification of optimal renal dosage adjustments for traditional and extended-infusion piperacillin-tazobactam dosing regimens in hospitalized patients. *Antimicrob Agents Chemother*. 2010;54(1):460-465. doi:10.1128/AAC.00296-09.
41. Sandri AM, Landersdorfer CB, Jacob J, et al. Population pharmacokinetics of intravenous polymyxin B in critically ill patients: implications for selection of dosage regimens. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2013;57(4):524-531. doi:10.1093/cid/cit334.
42. Baddour LM, Wilson WR, Bayer AS, et al. Infective Endocarditis in Adults: Diagnosis, Antimicrobial Therapy, and Management of Complications: A Scientific Statement for Healthcare Professionals From the American Heart Association. *Circulation*. 2015;132(15):1435-1486. doi:10.1161/CIR.0000000000000296.
43. Osmon DR, Berbari EF, Berendt AR, et al. Diagnosis and management of prosthetic joint infection: clinical practice guidelines by the Infectious Diseases Society of America. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2013;56(1):e1-e25. doi:10.1093/cid/cis803.
44. Berbari EF, Kanj SS, Kowalski TJ, et al. 2015 Infectious Diseases Society of America (IDSA) Clinical Practice Guidelines for the Diagnosis and Treatment of Native Vertebral Osteomyelitis in Adults. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2015;61(6):e26-46. doi:10.1093/cid/civ482.
45. Flanagan S, Minassian SL, Morris D, et al. Pharmacokinetics of tedizolid in subjects with renal or hepatic impairment. *Antimicrob Agents Chemother*. 2014;58(11):6471-6476. doi:10.1128/AAC.03431-14.
46. Nahata MC. Dosage regimens of trimethoprim/sulfamethoxazole (TPM/SMX) in patients with renal dysfunction. *Ann Pharmacother*. 1995;29(12):1300.
47. Rybak M, Lomaestro B, Rotschafer JC, et al. Therapeutic monitoring of vancomycin in adult patients: a consensus review of the American Society of Health-System Pharmacists, the Infectious Diseases Society of America, and the Society of Infectious Diseases Pharmacists. *Am J Health-Syst Pharm AJHP Off J Am Soc Health-Syst Pharm*. 2009;66(1):82-98. doi:10.2146/ajhp080434.
48. Liu C, Bayer A, Cosgrove SE, et al. Clinical practice guidelines by the infectious diseases society of america for the treatment of methicillin-resistant Staphylococcus aureus infections in adults and children. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2011;52(3):e18-55. doi:10.1093/cid/ciq146.
49. Cohen SH, Gerding DN, Johnson S, et al. Clinical Practice Guidelines for Clostridium difficile Infection in Adults: 2010 Update by the Society for Healthcare Epidemiology of America (SHEA) and the Infectious Diseases Society of America (IDSA). *Infect Control Hosp Epidemiol*. 2010;31(5):431-455. doi:10.1086/651706.
50. Koselke E, Kraft S, Smith J, Nagel J. Evaluation of the effect of obesity on voriconazole serum concentrations. *J Antimicrob Chemother*. 2012;67(12):2957-2962. doi:10.1093/jac/dks312.
51. Davies-Vorbrodt S, Ito JI, Tegtmeier BR, Dadwal SS, Kriengkauykiat J. Voriconazole serum concentrations in obese and overweight immunocompromised patients: a retrospective review. *Pharmacotherapy*. 2013;33(1):22-30. doi:10.1002/phar.1156.

A. Original Author/Date

Department of Pharmacy; 07/1998

B. Gatekeeper

Stanford Antimicrobial Stewardship Safety and Sustainability Program

C. Review and Renewal Requirement

This document will be reviewed every three years and as required by change of law or practice

D. Revision/Review History

Deepak Sisodiya, PharmD; 04/2005
Maggie Cudny, PharmD; 04/2007, 01/2009
Katherine Miller, PharmD; 01/2009
Sean Carlton, PharmD; 03/2010
Emily Mui, PharmD; 11/2010, 03/2011, 05/2012, 05/2013, 01/2014, 03/2017
Lina Meng, PharmD; 11/2010, 03/2011, 03/2017
Marisa Holubar, MD; 03/2017
Stan Deresinski, MD; 03/2017

E. Approvals

Antimicrobial Subcommittee 09/2004, 04/2007, 01/2009, 11/2010, 03/2011, 05/2012, 05/2013, 01/2014, 03/2017
Pharmacy & Therapeutics Committee 04/2007, 02/2009, 04/2010, 05/2011, 08/2012, 09/2012, 08/2013, 02/2014, 04/2017